

Predicting Well Performance and Failure Risk in Alberta Oil & Gas Wells



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Post Graduate Diploma in Data Analytics

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Project Summary

This project examines Alberta oil and gas well data to identify patterns in production, lifespan, and failure risk. Using Power BI and Python, it delivers a decision-support solution that helps detect stronger well profiles, spot early warning signs, and better understand long-term well performance.

Data & Method

- Integrated Alberta production and well-level data into an analytics-ready model.
- Applied filtering, feature engineering, and business-rule driven transformations.
- Built Power BI dashboards for production trend, lifespan, duration, and failure analysis.
- Used Python for forecasting, risk grouping, and early-performance analysis.

Key Findings

- Production volume trends upward over time, with recurring volatility.
- Failure-tagged wells are far fewer in count but show longer average duration, indicating many integrity issues occur later.
- Productive duration varies materially by well type, fluid type, and depth class.
- Wells can be segmented into long-lived, moderate-lived, and short-lived performance groups.

References

- AER General well data: <https://www.aer.ca/data-and-performance-reports/activity-and-data/lists-andactivities/general-well-data>
- AER Activity data: <https://www.aer.ca/data-and-performance-reports/activity-and-data/list-andactivities>
- AER Well casing failure report: <https://www2.aer.ca/t/Production/views/COMWellCasingFailureReport/WellCasingFailureReport.csv>
- Petrinex Production Data: <https://www.petrinex.ca/public-data/apd/>



Scan the QR to view the PowerBI Report



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Analytical Workflow



Production metrics and integrity signals are combined to support trend analysis, performance segmentation, and future well efficacy assessment.

9K

Failure-tagged wells

345K

Non-failure wells

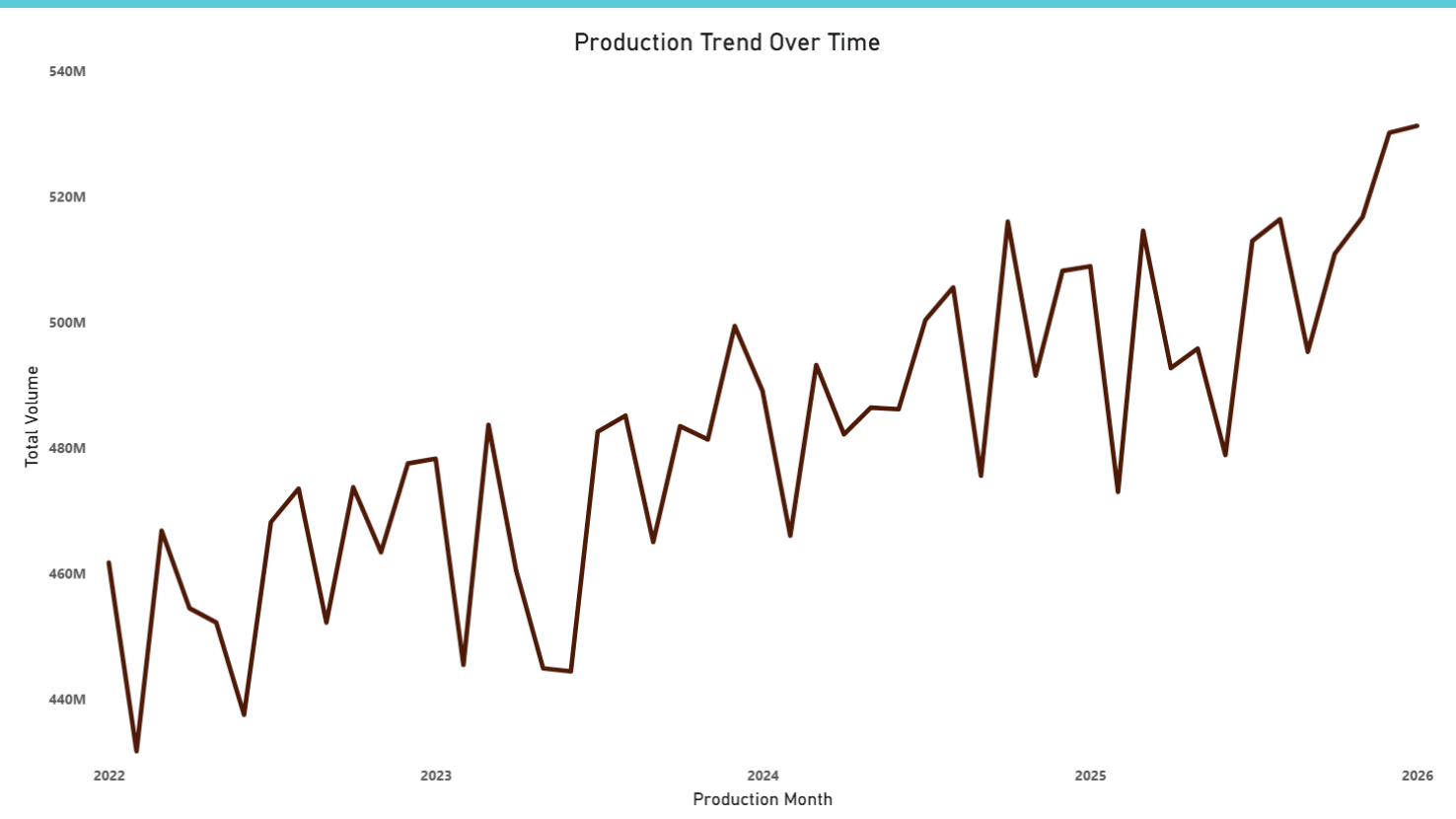
254.71

Avg duration – failure wells (months)

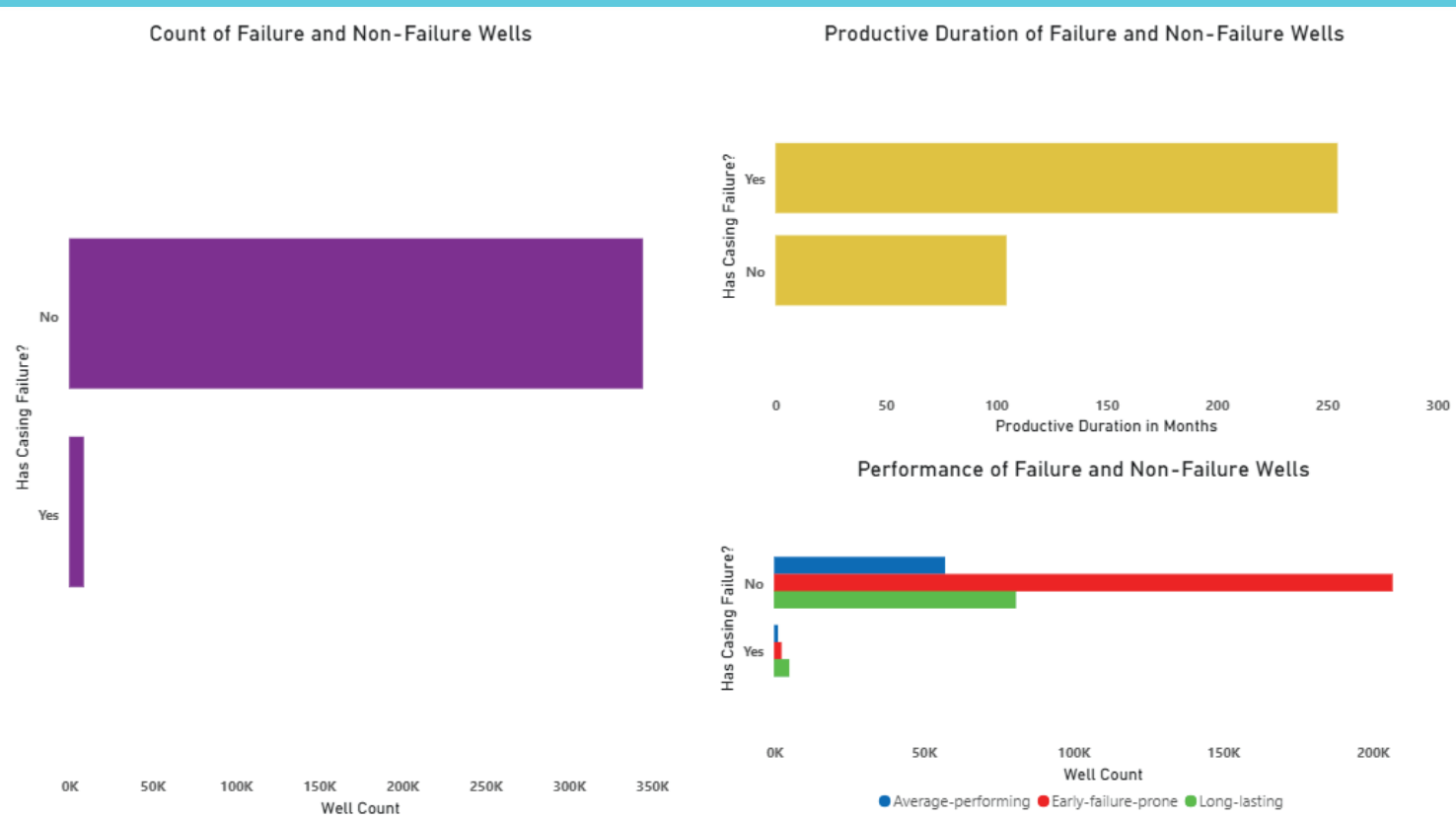
104.68

Avg duration – non-failure wells (months)

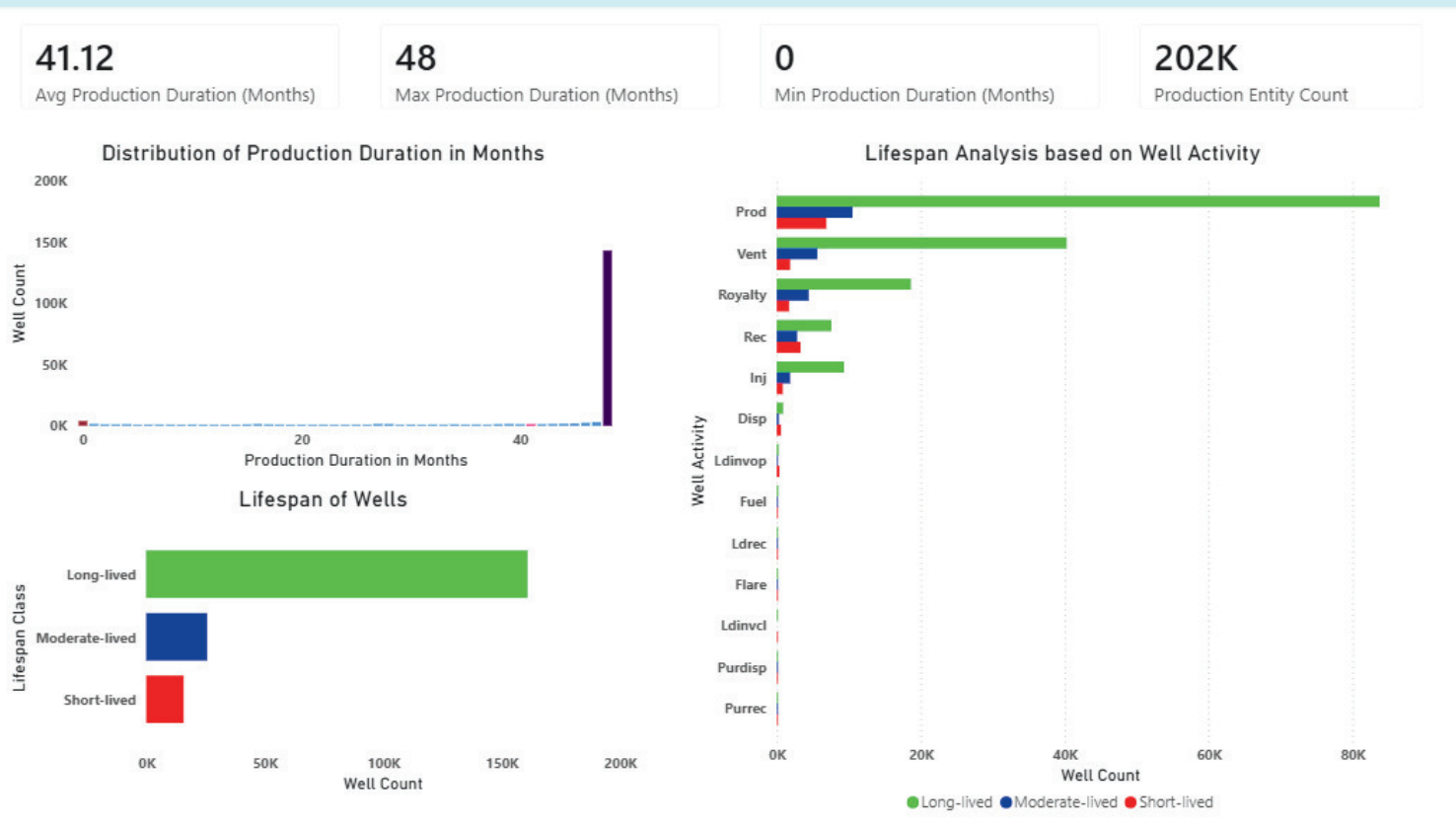
Production Trend



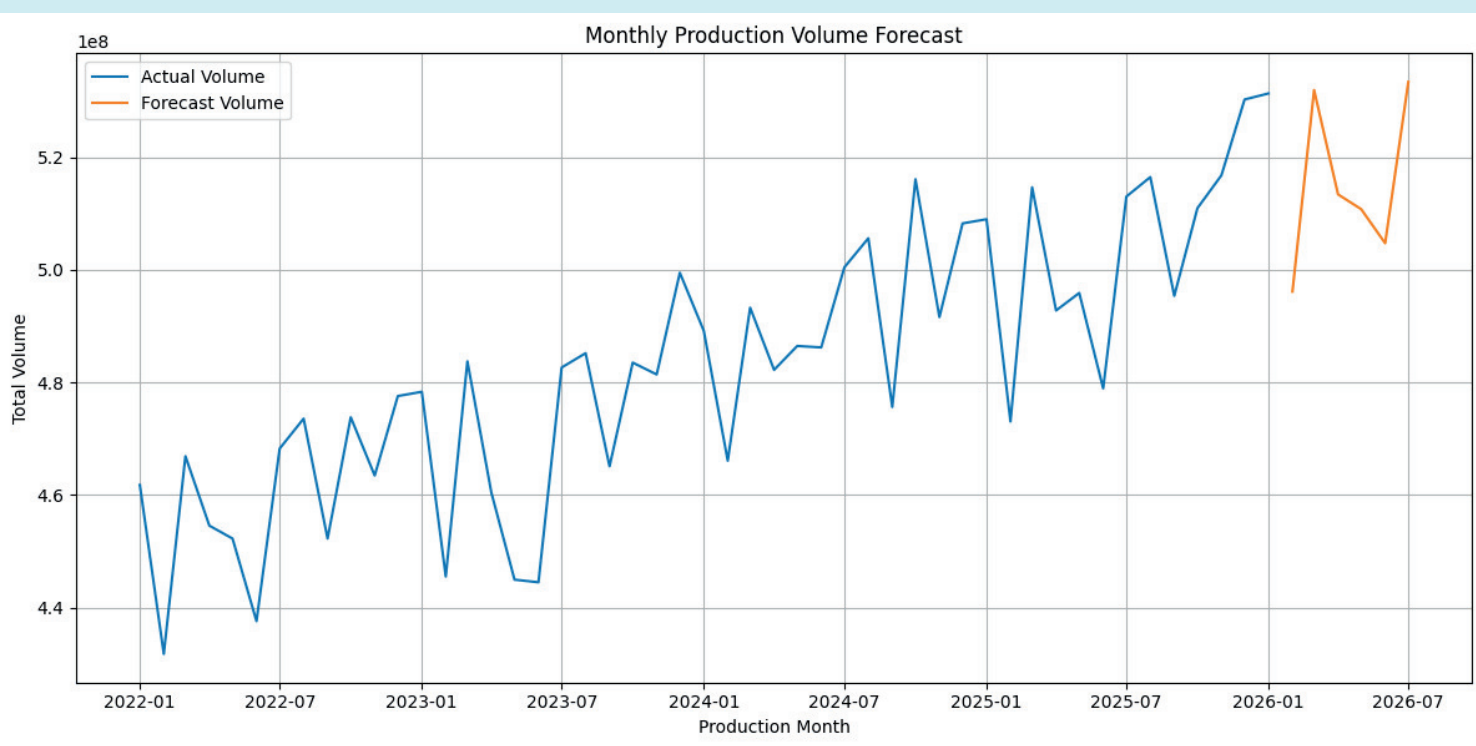
Failure vs Non-Failure Wells



Lifespan Analysis



Forecasted Production Volume



Productive Duration by Well Characteristics

